

Agency Cost of Free Cash Flow, Capital Allocation, and Payouts

by

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Abstract

Jensen's (1986) analysis of the agency costs of free cash flow radically transformed our understanding of corporate payout policy. This paper details the main pre-Jensen advances in the payout literature, explains how his analysis profoundly altered the way financial economists view payout policy, and discusses prominent regularities that provide real-world texture for understanding the importance of his insights about payout policy. These regularities include: (i) the dominance (measured as a percent of value or earnings) of the public equity markets by firms in the distribution phase of the corporate lifecycle; (ii) changes since the 1980s in aggregate cash payouts, the profitability of payers, payout rates, and the set of firms that dominate the payout supply; and (iii) the recent politicization of share repurchases.

Key words: capital allocation, free cash flow, payouts, politicization of finance.

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1. Introduction

Michael Jensen's (1986) paper "Agency Costs of Free Cash Flow, Corporate Finance, and Takeovers" radically advanced our understanding of corporate payout policy. Jensen's insights in this paper are surely among the most important contributions to the post-Modigliani and Miller (1958) literature on corporate financial policy. He presented his analysis compactly in an unrefereed paper just over six pages long. His insights about the importance of firms paying out free cash flow (FCF) emerged from his use of intuitive economic reasoning (rather than formal modeling) to make sense of the high-leverage takeovers and restructurings – especially of large oil firms – that were front-page news in the 1980s.

Jensen's message is that payouts matter because agency costs link them to the capital allocation process, with the value generated by a firm driven by the extent to which capital is invested wisely. When a firm has more resources than it can profitably invest, productive efficiency calls for the cash to be paid out. The reason is that retention of FCF in lieu of payout creates the opportunity for managers to destroy value by using excess resources in ways that serve their own self-interests.

This paper seeks to put in perspective Jensen's (1986) analysis of capital allocation and the importance of distributing FCF. We summarize the main pre-Jensen advances in the payout literature and explain how he profoundly altered the way financial economists view payout policy. We then highlight several empirical regularities about the payouts of publicly held firms to provide context for understanding the foundational importance of Jensen's contribution.

We find that firms that pay out cash (henceforth, payers) account for more than 85% of total market capitalization in all but two of the years between 1971 and 2023. Public equity markets in the U.S. are thus dominated by firms in the distribution stage of the corporate lifecycle, which are the focal point of Jensen (1986). The dominance of payers is even stronger in terms of earnings.

In Jensen's day, payouts were concentrated among a relatively small set of firms that generated the bulk of earnings. That remains true today. For example, the top 500 payers account for 90.0% of payouts in both 1986 and 2023 and for 85.6% and 92.8% of net income in those years, respectively. The top 500

payers account for about one-tenth of all firms (payers and nonpayers) in our CRSP/Compustat-based sample in 1986 and roughly one-eighth of all firms in 2023.

The identities of the largest payers have changed markedly since Jensen's day due to a shift in the set of public firms and industries that generate the bulk of earnings and to a decline in hostile takeover activity. GM, IBM, and GE – three of the most successful firms of the 20th century – were dominant payers in 1986 but are far less important today. In 1986, four of the 10 largest payers made the top ten because of one-time recapitalizations and/or large repurchase tender offers, with hostile takeover pressure visible in three cases. Such cases are largely absent today.

Technology firms, oil firms, and banks are now the most prominent contributors to the supply of payouts. Of these groups, only oil firms were major distributors of cash in Jensen's day. The rise of banks and technology firms reflects increased industry concentration (Kahle and Stulz (2017), Kwon, Ma, and Zimmerman (2024)) and the emergence of superstar firms (Autor et al. (2020), Ma, Pugsley, and Zimmerman (2024)). Growth by acquisition played a role in today's greater prominence of banks among the top distributors of cash and in the continued prominence of oil firms.

The payout mix has shifted away from dividends and toward share repurchases in the years since Jensen wrote his 1986 FCF paper, although dividends remain important. This shift was facilitated by the creation in 1982 of "safe-harbor" regulatory guidelines that enable a firm to repurchase shares without exposing managers to charges of attempts to manipulate the stock price.

The aggregate values of real payouts, real earnings, and real cash balances have increased markedly since the 1980s. Payout ratios (as a fraction of earnings) have increased materially since 2000, but there is no clear trend in the fraction of cash balances that firms typically pay out. Jensen's (1986) analysis can help explain why today's payouts are high, with the increases in the dollar amount of payouts and earnings-based payout ratios plausibly reflecting better monitoring due, e.g., to investor activism (Denes, Karpoff, and McWilliams (2017), Brav, Jiang, and Li (2022)) and to a general improvement in corporate governance (Holmstrom and Kaplan (2001)). Better monitoring also plausibly enabled the now much higher cash balances that firms typically hold by decreasing the likelihood that the cash would be wasted on bad

projects. The increase in cash balances cannot be explained by Jensen's (1986) analysis alone, which treats the absence of financial flexibility as a valuable means of limiting the ability of managers to spend corporate resources in self-interested ways.

The last 10 years or so have seen a surge in political attacks on corporate payouts. These criticisms focus on share repurchases, but dividends have also come under fire. The concentration, scale, and fungibility of cash payouts have made them attractive targets for politicians and social activists, who would prefer that firms fund causes they favor, including greater investment in workers, communities, innovation, and green energy.

Section 2 details the main pre-Jensen (1986) conceptual contributions to the payout literature. Section 3 discusses the main lessons Jensen (1986) taught us. Section 4 presents a compact descriptive portrait of the payout landscape that highlights the importance of Jensen's ideas as well as important changes since Jensen's day. Section 5 discusses the politicization of share repurchases. Section 6 summarizes two general theoretical principles that emerge from Jensen (1986) and draws on other work by Jensen to discuss his likely views of the politicization of payout policy.

2. A compact history of the payout literature before Jensen (1986)

In this section, we trace the main developments in the literature on payout policy that laid the foundation for Jensen's (1986) important contribution. Over the three decades leading up to the publication of Jensen's FCF paper, corporate-finance specialists wrestled with two ideas about payout policy. First, publicly held firms made large taxable dividend payouts to shareholders. Second, Miller and Modigliani (1961, MM) showed that, holding investment fixed and in the absence of taxes and other frictions, a firm's choice of payout policy is irrelevant. These ideas led researchers to search for a credible reason why firms would pay cash dividends when, absent taxes, the choice of payout policy doesn't matter. As we discuss in section 3, the incentive-based interplay between payouts and investment, which was ruled out by MM's fixed-investment assumption, turned out to be Jensen's critical insight.

Lintner's (1956) managerial survey was the first important contribution to the modern literature on payout policy. Lintner documented two regularities that have proved to be remarkably robust: (1) dividends are fundamentally connected to earnings, with managers believing that payouts should increase as earnings increase, and (2) managers are reluctant to cut regular dividends once initiated. In a rough sense, finding (1) anticipates the importance Jensen (1986) placed on FCF distribution because earnings drive what Wall Street refers to as free cash flow (operating cash flow net of capital expenditures). The definitional distinction is that Jensen uses FCF to refer to net cash flow under an *optimized* (value-maximizing) investment/operating policy, whereas Wall Street and corporate executives typically use the term to refer to operating cash flow net of actual capital expenditures.

Regarding (2), the post-Lintner literature contains strong empirical support for the idea that managers assign great importance to avoiding cuts in regular dividends; see, e.g., Brav et al. (2005). The robust support for regular dividends as a credible payout-commitment device weakens the case that Jensen (1986) made about debt playing a generally important role in forcing managers to disgorge FCF. We return to this issue in section 3.4 below.

Miller and Modigliani (1961, MM) fundamentally altered the prism through which corporate-finance specialists viewed the payout policies of firms whose shares are traded in large liquid stock markets. MM's main point was that dividend policy is irrelevant under frictionless conditions (no taxes, etc.) with investment/operating policy held fixed. One shortcoming of MM's analysis is that all payout variation occurs through changes in net stock issuance, which dictates that no incremental cash retention is allowed (DeAngelo, DeAngelo, and Skinner (2008, section 2.1)).

This limitation of MM's (1961) dividend-irrelevance proof is economically significant. Their formal analysis establishes the irrelevance of (1) equity-financed dividends, and (2) the dividend-repurchase mix, holding constant the dollar amount of cash distributed each period (at 100% of FCF every period). However, no one cared about either of these issues in the early days of the development of modern corporate-finance theory because, in real-world firms, equity-financed dividends and share repurchases were not empirically important.

Instead, the empirically relevant focus was squarely on the payout-versus-retention decision: How much should a firm distribute in the near-term as opposed to retain and reinvest to provide earnings to fund future payouts? MM's (1961) dividend-irrelevance proof is silent on what was unquestionably viewed as the main empirically relevant component of payout policy. Brennan (1971) and Rubinstein (1976) addressed this issue with analysis that allows a firm to reduce its current dividend payout and invest the retained cash in zero-NPV projects. In these models, payouts are lower in the current period, but the firm has more resources to fund larger future payouts due to the incremental investment, with no change in the overall value of the payout stream. Deferral of payouts thus entails no loss of value under frictionless conditions as long as the retained cash is invested in value-neutral ways and is eventually fully distributed.

This view of payout deferral inadvertently fostered serious and protracted confusion in the literature about the incentives to defer *taxable* payouts. If deferral of payouts (with reinvestment) entails no value loss for shareholders in the absence of taxes, why would a firm ever make a cash payout that would trigger an immediate tax liability? Deferral of taxable payouts would seem to be the optimal strategy. That intuition is the centerpiece of Black's (1976) famous paper, "The Dividend Puzzle." Here is what Black said (with emphasis added):

Under the assumptions of the Modigliani–Miller theorem, a firm has value even if it pays no dividends. Indeed, it has the same value it would have if it paid dividends.

If this theorem is correct, then a firm that pays a regular dividend equal to about half its normal earnings will be worth the same as an otherwise similar firm that pays no dividends and will never pay dividends.

In a world where dividends are taxed more heavily (for most investors) than capital gains, and where capital gains are not taxed until realized, *a corporation that pays no dividends will be more attractive to taxable individual investors than a similar corporation that pays dividends*. This will tend to increase the price of the non-dividend-paying corporation's stock. *Many corporations will be tempted to eliminate dividend payments*.

Black's argument also applies to share repurchases to the extent that they are taxable because they are priced to deliver a capital gain relative to the tax basis of the selling shareholder. The general claim, in a nutshell, is that *cash retention always dominates immediately taxable payout of cash*. In "Debt and Taxes," another highly influential paper, Miller (1977, pp. 266 and 270) invokes intuition advanced by Modigliani

and Miller (1958, fn 57) about the tax benefits of deferring payouts. Miller argued that it is reasonable to assume that the effective personal tax rate on equity is immaterial because long-term deferral of payouts will drive the present value of taxes on payouts close to zero.¹

In the tax-capitalization literature of public finance, Auerbach (1979) and Bradford (1981) pointed out that the existence of personal taxes on future payouts undermines the incentive to defer taxable payouts. This argument gained virtually no traction for two reasons. First, Merton Miller and Fischer Black were brilliant and revered figures and so few, if any, corporate-finance specialists were open to the idea they both might have been wrong in their views of payout deferral. Second, the tax-capitalization argument diminishes the incentive for firms to defer payouts but does not provide a fully convincing motive for firms to make immediately taxable payouts.

The literature's historical emphasis on the tax advantage of deferral also reflects a well-understood concern on the part of the government about tax-avoidance strategies, although Miller and Scholes (1978, p. 334) indicate that the historical record shows that the tax-code provisions designed to discourage corporate cash retention "pose no threat to the large, widely held corporations that actually pay most of the dividends." The important point is that the ostensible tax advantages of cash retention over immediate payout dominated the thinking of corporate-finance specialists in the 1970s.

Miller and Scholes (1978, MS) sought to rationalize immediately taxable dividend payments on the grounds that portfolio strategies using personal borrowing and insurance contracts could be used to convert dividend receipts into tax-deferred capital gains. Although MS's argument was taken seriously, it failed to sustain traction in the literature for two related reasons. First, their argument was not about rationalizing dividend payments *per se*, as it could be applied to personal interest income and capital-gains realizations, thus effectively turning any personal-tax-inclusive model into the MM corporate-tax framework – an

¹ There are problems with the reasoning of Black and Miller that were not identified for many years. They are detailed in DeAngelo, DeAngelo, and Skinner (2008, chapter 12). We ignore them here because our objective is to portray the state of understanding of payout policies at the time that Jensen (1986) was published. At that time, the view of the tax advantage of corporate retention advanced in "The Dividend Puzzle" and in "Debt and Taxes" was treated as state-of-the-art understanding in the corporate-finance literature.

approach to modeling the tax code whose empirical credibility Miller (1977) had recently undermined in “Debt and Taxes.” Second, as MS highlighted (p. 335), their tax-deferral strategy would, if employed by investors at scale, gut the personal income tax and effectively convert it into a consumption tax, which obviously has not happened.

Rozeff (1982) and Easterbrook (1984) argued that immediate payouts are beneficial because they force a firm to raise external capital more frequently, which fosters more effective monitoring by outside investors (because of the threat that they will supply funds at less attractive terms after detecting bad managerial behavior). This argument resonated with corporate-finance specialists because it was the first attempt to tie payout decisions to agency theory, which was in the process of transforming the field from the perfect-markets emphasis of MM (1958) to the framework of Jensen and Meckling (1976), which put the spotlight on contracting frictions and incentive problems.

At the same time, however, Rozeff’s (1982) and Easterbrook’s (1984) capital-market-monitoring argument was not widely accepted as a first-order motivation for firms to make taxable payouts. This assessment reflects the fact that it is hard to find examples of managers deliberately seeking to put themselves at the mercy of suppliers of capital, while there is ample indication that managers seek to insulate themselves from pressures associated with raising outside funding. See, for example, the study by Donaldson (1984) that Jensen (1986) references in his opening footnote.

Although the capital-market monitoring argument of Rozeff (1982) and Easterbrook (1984) identifies an economically sensible “on-the-margin” benefit of payouts, it is difficult to see that it, or for that matter any other argument advanced in the post-Lintner (1956) payout literature, provides a first-order affirmative motive for firms to distribute substantial amounts of cash in the near-term. That changed with the publication of Jensen (1986).

3. How Jensen (1986) changed our view of corporate finance

Jensen’s (1986) key insight is that *payout policy matters because of its connection to the capital allocation process and the incentives that govern the choice of a firm’s investment policy*. In this respect,

he built on the foundational insights of Jensen and Meckling (1976) and Myers (1977): When information and contracting costs are material, financial policy matters because managerial incentives forge a connection between the specific features of a firm's ownership and capital structure and the investment decisions chosen by managers. Jensen highlighted managerial opportunism as the factor that forged an empirically important connection between capital allocation and payouts. He advanced a compelling narrative about managers using cash retention (in lieu of payout) to fund investments that benefited them but destroyed value for shareholders.

Here, in Jensen's (1986) words, is the nature of the important relation between capital allocation and payouts that corporate-finance specialists had not yet appreciated:

Free cash flow is cash flow in excess of that required to fund all projects that have positive net present values when discounted at the relevant cost of capital. Conflicts of interest between shareholders and managers are especially severe when the organization generates substantial free cash flow. The problem is how to motivate managers to disgorge the cash rather than investing it at below the cost of capital or wasting it on organizational inefficiencies.

The main concern in Jensen (1986) is therefore getting excess cash out of the firm when there are significant risks of distortion in the capital allocation process. A secondary issue is the best way to do so, with the main options being regular dividend payments, special/extra dividends, share repurchases, and payments to holders of debt. Payments to debtholders are relevant here because, as Jensen (1986) emphasizes, debt is useful for forcing cash disgorgement when investors fear that managers will fail to distribute FCF through dividends and share repurchases. We discuss Jensen's view of the control function of debt in sections 3.4 and 3.5.

Although all these methods can alleviate the problem of potential FCF waste, total (corporate plus personal) tax payments will differ depending on the mix of dividends, share repurchases, and payments to creditors. Jensen treated these tax-related differentials as of second-order importance, and focused instead on mechanisms that pressure managers to send cash out of the firm.

Dividends and share buybacks entail direct cash payment to shareholders, which is what shareholders expect to receive at some point when they contribute capital to the firm. Jensen viewed the takeover market – and shareholder activism generally – as an important source of pressure on managers to deliver on this

expectation by distributing FCF to shareholders. The idea is that, if incumbent managers are over-retaining and wasting FCF, outsiders have incentives to replace them with new managers.

3.1 Jensen's (1986) theory: Two interpretations

The literature has two interpretations of Jensen's (1986) theory, with #1 nested as a special case of #2:

1. A theory that is important only for firms that lack profitable investment opportunities.
2. A theory that is important for firms generally, including those with high current earnings and liquidity relative to the current level of profitable investments.

#1 is based on the examples Jensen used in his paper to illustrate the importance of paying out cash. Those examples highlight managers' decisions to use current cash flow to fund new investments that are poor in an *absolute* sense. #2 is a natural – and arguably obvious – generalization of Jensen's discussion of managerial waste of available cash. It's not just firms with poor investment opportunities that have a potential agency problem with managers spending cash unwisely.

Firms with abundant cash *relative* to nontrivial positive-NPV investments face the same general agency problem, although it likely manifests differently than at firms that have few good investments. The problem is not simply empire building (or theft) by managers. Resource abundance can also foster an unduly lax managerial attitude about spending in general – e.g., project experimentation, adoption, and oversight – and that, in turn, can lead to value losses from suboptimal investment/operating policies. Such laxity can plausibly survive because of the real-world difficulties of measuring and evaluating the profitability of alternative investment/operating policies.

We focus on interpretation #2. It's not that #1 lacks empirical support. On the contrary, the literature contains evidence supporting #1. For example, Lang, Stulz, and Walkling (1991) report that firms with low Tobin's Q ratios – a proxy for poor investment opportunities – have stock returns that are significantly negatively related to cash flow when they announce tender offers to buy other firms.

The main reason we emphasize #2 is that #1 plausibly applies to a set of firms that is too narrow to explain the large aggregate cash payouts that have been observed in publicly held firms for many years (see section 4 below). It seems unlikely that a pervasive absence of good investment opportunities at firms

across the board can explain the magnitude of aggregate payouts. On the contrary, flourishing technology firms like Apple and Microsoft plausibly have ample good new investments that are exceeded by their even larger earnings from previously undertaken investments.

Our decision to emphasize interpretation #2 also reflects Jensen's (2005) discussion of the agency costs of overvalued equity (ACOE). Jensen's ACOE argument is a straightforward generalization of #2 that recognizes a different source of cash abundance that can lead managers to make unwise investments: The ability to issue shares that are significantly overpriced by the stock market.

3.2 The sense in which Jensen (1986) is a foundational element of corporate-finance theory

In their pioneering work, MM (1958, 1961) insisted on maintaining a sharp conceptual distinction between a firm's investment and financing decisions. In the MM framing of corporate finance, that distinction effectively dictates that all value is generated through investment *per se* and that financing decisions have no impact on the value of the firm or shareholder wealth.

There are two important respects in which Jensen's contribution should be viewed as foundational to the way financial economists approach issues of corporate finance. One involves the link between capital allocation and payouts and the other involves the basic building blocks of corporate-finance theory. Let us take these two points in turn.

Although managerial waste of FCF was central to the case that Jensen (1986) articulated, the more general issue concerns the critical linkage between capital allocation and payout policy. That linkage is present even when managers act in shareholders' best interests, as generating maximum value for shareholders involves the (eventual) payout of cash that cannot be profitably reinvested.

The framing of corporate-finance theory usually begins with Modigliani and Miller's (1958) simple two-security (debt and equity) model of the firm. It then adds various frictions associated with the use of debt and equity financing and solves for a capital structure optimum as a tradeoff of the posited frictions.

One could instead start with a firm that has a single security class (equity) and operates in a frictionless world, and then examine two broad categories of decisions, each with its own set of frictions, and how key frictions in each category affect the use of debt and equity financing. The first broad category entails

corporate *access to external funding*, for which the asymmetric information analysis of Myers and Majluf (1984) is the foundational contribution. The second broad category entails *getting cash distributed by the firm to the suppliers of capital* after earlier rounds of funding have been allocated to investment and the rewards from investment realized, for which Jensen's (1986) FCF analysis is the foundational contribution.

Viewed from this perspective, the debt-versus-equity capital structure choice is not the central or foundational problem of corporate finance. Instead, corporate financial policy is driven by two principles: (i) getting cash into a firm in the most efficient way to finance productive investments, and (ii) distributing the rewards from investment to the suppliers of capital so that it is not wasted. The two principles are tightly linked, as textbook-level logic about share valuation indicates: Firms will be able to raise capital at attractive terms only to the extent investors expect they will eventually receive payouts that have a present value as least as large as that of the capital they contributed.

This concern about the linkage between capital contributions and expected future payouts is present in every financing scenario. It exists in the simplest scenario in which a firm has a single outstanding security class: Equity or common stock. Starting from that baseline, we can think of debt (and other securities and contractual mechanisms) as tools firms use to deal efficiently with the various frictions that are relevant for the interdependent capital-infusion and distributional components of financial policy that are the focus of Myers and Majluf (1984) and Jensen (1986) respectively.

Importantly then, payout policy should not be viewed as subordinated in importance to a firm's debt-equity mix, which is how Miller and Modigliani's (1961) dividend policy follow-up to their 1958 capital structure paper encouraged us to think. Rather, Jensen's (1986) analysis has helped us see that payout policy is a primary building block of corporate financial policy central to value creation.

This framing of corporate-finance theory illustrates how Jensen's (2005) analysis of the Agency Costs of Overvalued Equity (ACOE) is the conceptual complement of his FCF paper. The FCF paper is concerned with managerial warping of investment decisions facilitated by over-retention of cash while the ACOE paper is concerned with the same type of problem generated by infusions of cash made possible when the market overvalues a firm's shares.

This alternative framing of theory also leads to an empirically more relevant approach to understanding capital structure and cash dynamics. The MM framing naturally leads to friction-inclusive tradeoff models of capital structure as in Robichek and Myers (1966, RM). RM-style tradeoff models all have a target leverage optimum and counterfactually predict that every proactive capital structure change moves a firm toward its target. When theory is instead grounded in principles (i) and (ii), debt and cash are used as transitory funding vehicles, with firms issuing debt and drawing down cash balances to meet unanticipated funding needs and then deleveraging (repaying debt and rebuilding cash balances) as earnings realizations permit. That is a far more realistic portrait of debt and cash dynamics than RM-style tradeoff models predict. See the capital structure evidence cataloged in DeAngelo (2022, section IV).

3.3 A lifecycle theory of corporate financial policy

This re-framing of basic theory leads immediately to a predicted linkage between the scale of payouts and FCF generation over a corporate lifecycle and, more generally points to a lifecycle theory of firms, as Jensen (1993) emphasized in his AFA Presidential Address. The lifecycle payout-FCF linkage is obvious from the observation that young growth firms that are not yet generating FCF require external funds to pursue their investment opportunities, and so are poorly suited to distribute large amounts of cash. In contrast, mature firms plausibly have generous payout policies because they generate more cash than they can profitably invest. The implication is that growth firms should focus largely on cash infusions and make few payouts while mature firms should be distributing the FCF they generate.

The less obvious implication is Jensen's (1993, section IV) observation that there is a fundamental asymmetry in the behavior of firms operating in the growth and decline stages of the corporate lifecycle. These different stages call not only for differences in capital raising and distribution, but also for fundamentally different management skills to navigate the different operational challenges that each stage presents. A discussion of the latter point is beyond the scope of this paper. We raise it here to highlight that an important general theme – one traceable to Jensen (1986) – connects many of Jensen's important insights about corporate financial policy.

3.4 The control function of debt versus the value of financial flexibility

Jensen viewed a firm's payout decisions as inherently linked to its capital structure through what he called the control function of debt. The premise is that, with rare exception, debt creates a legally enforceable obligation to pay cash out of the firm to claim holders. The control function idea is that debt can be useful for forcing cash disgorgement when (i) managers are effectively reneging on their implicit contract to distribute FCF to shareholders (say with unduly low regular dividends) and (ii) pressures from activist shareholders are inadequate to prevent excess retention. Jensen's point is that, in such situations, debt is a vehicle for using the contract-enforcement power of the state as a device for disciplining managers to ensure that FCF is paid out.

There is clearly merit in the control function of debt idea. However, in our judgment, Jensen put too much emphasis on this factor as a driver of corporate capital structures. Yes, he recognized that increased leverage also comes with agency and bankruptcy costs and that such costs attenuate the incentive to use debt to motivate managers to distribute FCF. But he ignored the benefits of financial flexibility. Indeed, he treated the absence of financial flexibility as an unambiguously positive feature, not a bug, because debt forces managers to pay out rather than waste firm resources. Although Jensen didn't discuss cash balances in his FCF paper, the spirit of his analysis indicates higher net-debt obligations (debt minus cash holdings) are desirable because of the discipline they impose on managerial decision-making. Corporate-finance specialists now recognize that there are counter-vailing economic forces at work that encourage firms to have more conservative financial policies, as manifested in lower leverage ratios and/or higher cash balances, and as evidenced, e.g., by higher credit ratings.

The conceptual point that is missing in Jensen (1986) is recognition that there is value in untapped debt capacity and/or large cash balances because they provide low marginal cost access to resources to meet imperfectly anticipated funding or liquidity needs. This idea was made in convincing fashion in Myers and Majluf's (1984) breakthrough analysis of financial policy dynamics, although Higgins (1972) previously discussed the value of corporate cash retention in anticipation of future funding needs.

Hindsight is 20-20 but it is clear from today's vantage point that Jensen's (1986) discussion would have benefited from recognizing the value of financial flexibility and explicitly considering investment and payout decisions in a dynamic context. His discussion is framed in the spirit of a "one-shot" decision model, as in standard tradeoff theories of capital structure that follow the modeling template of Robichek and Myers (1966).

The free cash *flow* language encourages readers to think narrowly about decisions made in a single period about whether to distribute or retain and immediately invest profits. A dynamic framing would have recognized that cash retention today can fund immediate investment or can stockpile internally available highly liquid resources to fund future investment outlays. It would have shifted emphasis away from a flow-based measure of waste potential and put the spotlight on the extent of cash balances or some other such measure of liquid asset holdings that provide managers with opportunities to make distortionary firm investments that serve their narrow self-interests.

This sort of dynamic framing of Jensen (1986) would have led immediately to the idea that, for all firms, there is an inherent tradeoff between preserving financial flexibility (to meet imperfectly anticipated funding and liquidity needs) and having managers operate on a tight financial leash with strong pressures to distribute cash and not maintain either large cash balances or material untapped debt capacity. In short, this approach would have provided an empirically more powerful framework that combined (i) the narrow "one-shot" FCF waste problems central to Jensen (1986) with (ii) the financial flexibility considerations central to the dynamic framing of corporate finance in Myers and Majluf (1984).

The key empirical problem for the narrow way Jensen (1986) framed his argument is that it is easy to find firms that distribute large amounts of cash through stock repurchases and dividends despite having ample financial flexibility with high cash holdings and conservative leverage ratios. See, for example, the discussion below of the cash payouts of prominent technology firms like Apple and Microsoft. These firms make large cash distributions while maintaining substantial cash holdings and facing no debt-related pressures to make large payouts.

3.5 What do LBOs tell us about the importance of the control function of debt?

Jensen's (1986) assertion that debt plays an important disciplinary role (in mitigating FCF waste) gained traction in the literature because of his seemingly convincing claims about the motives for using debt in leveraged buyouts (LBOs):

Many of the benefits in going private and leveraged buyout (LBO) transactions seem to be due to the control function of debt....

Desirable leveraged buyout candidates are frequently firms or divisions of larger firms that have stable business histories and substantial free cash flow (i.e., low growth prospects and high potential for generating cash flows) – situations where agency costs of free cash flow are likely to be high.

Although Jensen's portrayal of the characteristics of many LBO candidates is correct, those characteristics do not conclusively establish a motive for LBO firms to be highly leveraged *due to the control function of debt*. There are two issues here. The first is that firms acquired in an LBO typically have concentrated equity ownership and boards of directors dominated by representatives of the private-equity firm(s) that supplied the bulk of the equity funding to make the acquisition. In terms of incentives, these post-acquisition ownership and governance features make the LBO firm close in substance to the single owner-manager case in Jensen and Meckling (1976) where agency costs are zero (and managers do not waste FCF). That means that debt is largely a redundant control device for reducing FCF waste in firms that have undergone an LBO.

The second issue is that heavy reliance on debt in LBOs could reflect other factors. For example, it could reflect the attraction to prospective lenders of borrowers with strong and reliable cash flows. It could also reflect a desire by private-equity firms to treat their pool of investable equity capital as a scarce resource while capturing tax benefits from debt financing.

Our point here is not that Jensen's (1986) control function of debt argument is wrong. Without a doubt, debt sometimes makes good sense as a disciplinary device to mitigate FCF waste, especially when managers have lost credibility with investors and regular dividend payments are not perceived as a strong commitment to distribute FCF. Our point is that Jensen's argument is not convincing of the *pervasive* importance of the control function motive for firms that have undergone LBOs to have large amounts of debt.

The idea that LBOs are a means of addressing the FCF waste problem central to Jensen (1986) *is* compelling, however. The target firm’s now-high concentration of equity ownership and the presence of private-equity owners on the board of directors create stronger incentives for managers to make wise capital allocation decisions than under dispersed public ownership.

4. Prominent empirical features of the payout landscape

This section provides a descriptive portrait of the payout policies of publicly held firms that illustrates the importance of issues that Jensen (1986) analyzed and that highlights several notable changes in the payout landscape since Jensen’s day. While some of our analysis is novel, to a large degree we build on empirical work in prior studies. We analyze all firms in the CRSP/Compustat universe over 1971-2023. While prior work tends to exclude financial firms and utilities, we include those firms to provide a comprehensive view of payouts. Appendix A details our sampling criteria and data definitions.

4.1 Market value, net income, and aggregate dividend and share-repurchase payouts

Figure 1 shows that public equity markets in the U.S. are dominated by firms that pay out cash. For almost two-thirds of the years from 1971 to 2023, the market capitalization of payers exceeds 90% of total market capitalization, falling below 85% in just two years (when unsustainably high stock prices for young tech firms temporarily reduced the market capitalization of payers to around 80% of the total).

The dominance of payers is even more pronounced when we look at earnings. Figure 2 reports earnings measured as net income (panel A) and operating income (panel B). These figures show that nonpayers contribute little or nothing to aggregate earnings. Some nonpayers, of course, generate earnings, but the losses reported by others are usually large enough to offset all such earnings and, in a few years are so large as to drive the aggregate earnings of nonpayers well into negative territory.

Figure 2 also shows that the inflation-adjusted aggregate earnings of publicly held firms that pay out cash is substantial in every year from 1971 to 2023 and has become much larger in the last two decades. For example, aggregate real net income essentially doubled from just below \$1 trillion around the turn of the century to around \$2 trillion in 2023.

Figure 3 shows that the growth in aggregate earnings in the 21st century has been accompanied by a comparably impressive increase in aggregate real payouts by publicly held firms. The figure shows that dividend payments increased markedly over the full period 1971-2023 and especially so over the last 20 years. The surge in share repurchases over the last 20 years or so has been even greater, with repurchases now accounting for more than half of aggregate payouts by publicly held firms. Increases in aggregate payouts and the impressive growth in share repurchases have been documented in many prior studies; see, e.g., Kahle and Stulz (2021) for a recent tabulation.

Although real dividend payments have grown consistently since 1971, there has been substantial year-to-year variation in the number and percent of firms that pay dividends (not reported in Figure 1). Fama and French (2001) document a sharp decrease in the fraction of firms that pay dividends from 1978 to 1999. Dividend payments, however, were not disappearing. Larger total dollars of dividends were being delivered by a smaller number of payers, with growth-by acquisition playing an important role in the consolidation of the dividend supply; see DeAngelo, DeAngelo, and Skinner (2004).

The resilience of dividends is an indication of the importance of the issues that Jensen (1986) emphasized. Although repurchases have tax and flexibility advantages over dividends, regular dividends credibly convey managers' commitment to distribute FCF rather than retain and fritter it on dubious projects and acquisitions. This commitment is evident in (1) the large scale of dividend payments (per Figure 3), and in (2) the asymmetric nature of dividend changes: Dividend cuts are rare and large (and occur mostly in situations of financial distress) while dividend increases are small, frequent, and regular (some firms have increased dividends for decades; e.g., Coca-Cola has increased its quarterly dividend every year since the early 1960s and paid a dividend each year since the 1920s). These findings are also borne out in managerial surveys; see, e.g., Brav et al. (2005).

Figure 4 partitions the annual payouts from Figure 3 into those made by financials, utilities, and industrials (defined as all nonfinancial and nonutility firms). In all years, industrials are by far the largest supplier of payouts, although the payouts of financial firms have grown over time and are now substantial in their own right. Utilities make only minor contributions to the aggregate supply of payouts.

Aggregate payouts by publicly held firms are consistently much larger than the equity infusions provided through SEOs and IPOs – a regularity that is evident by comparing our Figures 3 and 4 to the equity infusions amounts for 1990-2023 reported by Huang, Ritter, and Zhang (2024, HRZ). For the median year over this period, total SEO+IPO infusions are 12.1% of aggregate payouts, and reach a maximum of 36.2% in the year 2000 (during the hot-issue market of the tech-stock-bubble period).²

4.2 The corporate-payout cross-section in Jensen’s day versus today

Table 1 juxtaposes key features of the payout cross-section in 1986, the year Jensen’s FCF paper was published, against those for 2023, the most recent year with data available. The number of payers is roughly the same – 2,904 in 1986 versus 2,999 in 2023 – but the fraction of payers is substantially higher today (73.3% versus 54.8%).

For many years, payouts have been concentrated among a relatively small set of firms that generate the bulk of earnings – a signature feature of the payout landscape evident in both 1986 and 2023. In both years, the top 500 payers account for 90.0% of aggregate payouts while the top 10 account for about 20.0% (panel A of Table 1). The top 10 and top 500 payers respectively generate 25.0% and 92.8% of aggregate earnings in 2023 – up from 14.9% and 85.6% in 1986 (panel B). The increase for the top 10 is a bit misleading because, as noted below, three of the 10 largest payers in 1986 made large one-time payouts in response to visible threats of hostile takeover. Firms that distribute cash in 1986 and 2023 respectively account for 92.5% and 92.3% of total market capitalization (panel C of Table 1).

Consistent with Figure 3, from 1986 to 2023 we observe substantial increases across the board in both the fraction of payouts made through share repurchases (panel D of Table 1) and the fraction of payers that repurchase (panel E). While the percent of payers that pay dividends has declined from 81.4% to 59.1%, this decline is much more muted among the largest payers (panel F) which says that dividends remain important, especially for the largest payers.

² HRZ’s data are equity proceeds before the payment of banking fees and do not include IPOs by banks and S&Ls and issuances by OTC firms. The latter considerations are minor measurement details and do not alter the basic inference that aggregate payouts consistently swamp total SEO+IPO infusions for publicly held firms.

Table 2 reports key features of the 10 firms that made the largest total real payouts over the five-year periods 1982-1986 (panel A) and 2019-2023 (panel B), as well as in 1986 (panel C) and 2023 (panel D). We emphasize the five-year profiles because a firm's distributions (and earnings) in a single year are sometimes affected by unusual events. The latter point will be clear from our discussion of panel C below.

A dramatic shift in the identity of the largest payers from Jensen's day is evident when we compare panels A and B of Table 2. Five of the top 10 payers over 1982-1986 are oil firms while only one (IBM at #3) is a technology firm. Five of the top 10 payers over 2019-2023 are technology firms and all are much more important to the payout supply than IBM (which has fallen to #46). The more recent period's top 10 includes three large, systemically important banks (all new to the highly ranked set of payers) and two oil firms (both prominent payers in the 1982-1986 period).

ExxonMobil is the only firm to make the top 10 in both periods. The decline from panel A to panel B in the number of oil firms is a bit misleading because the industry underwent massive consolidation through mergers in the years since Jensen's day. Mergers account for the disappearance of Atlantic Richfield (#4), Conoco (#6), Amoco (#7), and Texaco (#10). In panel B's top 10, the two oil firms – Exxon Mobil (#7) and Chevron (#10) – grew substantially through acquisition. Growth by acquisition also played a role in the emergence of banks among the most prominent distributors of cash in recent years.

Among the top 10 payers, the real dollar magnitudes of payouts and earnings in recent years are far greater than in Jensen's day. For example, over 2019-2023, Apple (ranked #1) paid out a bit over half a trillion inflation-adjusted dollars while reporting real earnings of just under that amount. In real terms, Apple's payout is almost 10 times the cumulative real payouts of Exxon over 1982-1986, which ranked #1 for that period. Chevron, the #10 firm over 2019-2023, would have been by far the #1 cash distributor over 1982-1986 if it had made its recent real payouts of \$90.9 billion in that period.

For the top payers, dividends have become less important as a percent of total payouts but remain substantial. Eight of the top 10 payers over 2019-2023 paid dividends including the top five dividend payers. The other two firms – Alphabet and Meta – initiated regular dividends in 2024, suggesting that dividends remain a pervasively important payout vehicle, at least for prominent firms. Although

repurchases now deliver more cash to shareholders, dividends remain a significant part of the payout landscape. The ongoing magnitude of dividends and the implicit commitment inherent in regular dividends – a commitment that is not present with repurchases and extra/special dividends – suggests they play a nontrivial role in mitigating the resource-waste problem central to Jensen (1986).

The data for 1986 and 2023 in panels C and D of Table 2 support the inferences we have drawn from panels A and B. The important new observation is that, in 1986, four of the top 10 firms made the list because of one-time payouts, three of which were motivated by visible hostile takeover activity. Such activity was regularly front-page news in the 1980s and loomed large in the way Jensen came to the ideas in his landmark paper on payout policy – a reality he highlighted by including the word “takeovers” in the title of his 1986 paper.

Table 2 further shows remarkably high cash-balance ratios in 2023 for most of the top payers (panels B and D, far right column) – a fact that is especially notable for technology giants Apple, Alphabet, Microsoft, and Meta. Each of these firms distributed a large fraction of their earnings over 2019-2023 (panel B) and in 2023 alone (panel D). Their high payout ratios are misleading in the sense that they are not indicative of managers having limited ability to spend the firm’s liquid resources on unwise investments that serve their narrow self-interest. On the contrary, these firms maintain large reservoirs of cash and marketable securities. Those reservoirs would seem to give managers ample opportunity to make unwise investments, which was Jensen’s (1986) central concern. This phenomenon is not unique to these large technology firms, as we next document.

4.3 Payout rates relative to earnings and cash balances

Figure 5 reports payout ratios from 1971 through 2023 for all payers with positive earnings in a given year. Panel A reports the ratio of payouts to operating income (OI), while panel B reports the ratio of payouts to net income (NI). The dashed line shows the sum of all firms’ payouts divided by the sum of their earnings. The solid line documents median earnings-payout ratios. The former is more representative of firms that have higher earnings and make larger payouts.

The signature empirical regularity in Figure 5 is that, in recent years, firms tend to pay out a larger fraction of earnings than they did 25 years ago. For both payout ratios, the aggregate and median ratios are roughly 0.2 greater in recent years than they were around 2000 – firms now tend to distribute about 20% more of earnings than they did around the turn of the century. This increase in the rate at which earnings are distributed is very close to the increase since the year 2000 documented by Kahle and Stulz (2021, figure 2) for industrial firms.

Holdings of cash (and marketable securities) for payers have increased substantially since 2000. Figure 6 indicates that aggregate real cash balances for payers were around \$3 trillion in 2000 compared to numbers in excess of \$7 trillion in recent years. Figure 7 shows that these larger holdings of cash translate to roughly a 50% increase (from 0.08 to 0.12) in the aggregate ratio of cash-to-total assets and to roughly a 100% increase in the median payer’s Cash/TA ratio (from 0.04 to 0.08) since the turn of the century. This recent increase in Cash/TA ratios is a continuation of a trend for industrial firms from 1980 to 2006 documented by Bates, Kahle, and Stulz (2009).

Thus, although firms now tend to pay out a larger fraction of earnings, many do so while holding considerably more cash. The secular increase in payout ratios in Figure 5 therefore does not necessarily say that payout policies have become more attentive to agency costs of free cash flow, which is Jensen’s (1986) central concern. The reality instead, as Figures 6 and 7 document, is that firms now tend to operate with greater cash holdings, which indicates that managers have greater command over resources that they could deploy in self-interested, unproductive ways.

The natural question then becomes: To what extent has there been a change in the rate at which firms draw down their (now much greater) cash holdings through payouts to shareholders? To address this question, we calculate a cash-balance-based payout ratio, defined as the dollar payout in a specified year divided by the sum of cash payouts and the year-end cash balance (i.e., the post-payout holdings of cash). Figure 8 reports an aggregate version of this ratio (the sum of all firms’ payouts divided by the sum of all firms’ cash balances and payouts) and the median firm’s ratio for each year in question.

Figure 8 indicates that the aggregate and median cash-balance-based payout ratios both largely remain in the range 0.015 to 0.200 from the year 2000 to the present. The aggregate cash-balance-based payout ratio remains within that same range back to the early 1980s, while the ratio for the median firm largely remains within the range 0.200 to 0.250 from the early 1970s to the early 1990s. There is no sign in these data of a secular upward trend in cash-balance-based payout ratios.

We conclude that firms have not become more aggressive in recent years in the rate at which they typically pay out cash. Although the data indicate that payouts have trended higher relative to earnings, that trend has been accompanied by a substantial increase in cash holdings. The upshot is that the higher payout ratios are occurring against a backdrop of now much larger cash holdings. Firms have not transitioned to operating on a tighter financial leash, as would be the case if Jensen's (1986) logic were the main driver of the trends documented in Figures 5 through 8. Instead, the trend has been toward firms operating with greater financial flexibility (as indicated by larger holdings of cash), which provides a cushion to support their now-higher earnings-based payout ratios.

That is not to say that Jensen's (1986) logic is unimportant for understanding the large scale of observed payouts by publicly held firms. On the contrary, the large scale of firms' cash balances provides a plausible basis for concluding that his logic is of first-order importance for understanding the scale of payouts. Consider the counterfactual scenario of firms having retained rather than distributed large amounts of cash. Their large stockpile of cash that is evident in Figure 6 would have been trillions of dollars greater, as can be seen from the annual scale of payouts reported in Figure 4. The result of this hypothetical additional retention (in lieu of payout) would be to create a scale of liquid-resource abundance within firms that seems far out of proportion to any reasonable assessment of their profitable near-term investments.

4.4 FCF and other explanations for corporate payouts

The literature contains three broad factors that potentially explain decisions to pay out cash:

- (1) Free cash flow (FCF) distribution, with an emphasis on the agency costs that arise when managers have opportunities invest a firm's liquid assets in self-interested, unproductive ways.

- (2) Other motives that receive major emphasis in the literature and finance textbooks: Signaling future earnings to outside investors, meeting the portfolio demands of specific investor clienteles for stable cash distributions, and the exploitation of stock-market-timing opportunities.
- (3) Other more nuanced (and sometimes opportunistic) motives such as: (i) use of share repurchases to increase managerial bonuses that depend on EPS, (ii) use of repurchases to boost EPS and/or meet quarterly EPS expectations, (iii) use of repurchases to offset the dilutive effects of employee stock options, (iv) dividend avoidance because the exercise price on employee stock options are not dividend-protected, (v) share repurchases to increase the pay-performance sensitivity of equity held by managers, (vi) share repurchases that deter takeovers by buying the shares of investors most willing to sell to a bidder, thereby concentrating ownership among non-selling shareholders, and (vii) dividend and repurchase payouts to increase leverage in the direction of an ostensible target ratio.

The literature on corporate payout policy is massive and a thorough review is well beyond what we can reasonably provide in this paper. For readers interested in detailed discussions of the evidence on the importance of these factors, see Allen and Michaely (2003), Vermaelen (2005), Kalay and Lemmon (2008), DeAngelo, DeAngelo, and Skinner (2009), Farre, Michaely, and Schmalz (2014), Bonaime and Kahle (2024), and Leary and Nukala (2024).

In our view, the data we summarize above provide a compelling basis for FCF distribution to be the foundational element of any empirically credible theory of payout policy. Aggregate payouts are simply too large, and the payouts of individual firms are too tightly linked to earnings, to be plausibly explained by the other factors in (2) and (3). Consider, for example, the large cash distributions and earnings of Apple reported in panels B and D of Table 2. Apple has many ways of communicating with investors about its future earnings and has a track record to make its disclosure statements credible, and so simply does not need to distribute billions of dollars to send a convincing signal about future profits to investors. See, for example, Lu and Skinner (2024), who discuss how managers use guidance in conjunction with payouts to convey information about their firms' prospects.

In addition to the question of how much cash to pay out, the literature frequently asks: What explains the dividend-versus-repurchase mix? In terms of Jensen's (1986) analysis, regular dividends entail a stronger implicit commitment of continued distribution of cash. Many of the other factors in (2) and (3) also plausibly exert influence on the dividend-repurchase mix. As with the level of payouts, providing a detailed assessment of the empirical importance of these other factors is well beyond the scope of this paper.

5. The political targeting of share repurchases

Throughout his career, Jensen viewed political interference in corporate decision-making as a serious threat to the efficient functioning of the free-enterprise system. Jensen (1991) spoke out strongly against attempts by managers, organized labor, and politicians to influence laws and regulations that would benefit incumbents by impeding activist investors who sought to use the market for corporate control to discipline firms' capital allocation. Jensen's analysis of activist investors in the corporate-control market of the 1980s played a central role in helping him come to the insights in his 1986 paper on the importance of pressuring managers to distribute a firm's free cash flow to investors.

As Jensen (1986) emphasized, in the 1980s, oil firms faced pressure from investors to increase cash payouts and curtail investment. While such pressures are surely present today, oil firms in recent years have also faced pressure from politicians and environmental activists to reduce their "large" payouts and instead use the retained cash to invest in green energy and benefits for their employees; see, e.g., Maloney and Khanna (2022) and Natural Resource Defense Council (2023).

Today's political targeting of payouts occurs broadly across industries and focuses mainly on criticisms of share repurchases, although dividend payments have also been targeted. For example, in a 2019 *New York Times* op-ed titled "Schumer and Sanders: Limit Corporate Stock Buybacks," Senate Majority Leader Chuck Schumer and Senator Bernie Sanders state:

.... corporate boardrooms have become obsessed with maximizing only shareholder earnings to the detriment of workers and the long-term strength of their companies, helping to create the worst level of income inequality in decades.

One way in which this pervasive corporate ethos manifests itself is the explosion of stock buybacks.

....Americans should be outraged that these profitable corporations are laying off workers while spending billions of dollars to boost their stock's value to further enrich the wealthy few. If corporations continue to purchase their own stock at this rate, income disparities will continue to grow, productivity will suffer, the long-term strength of companies will diminish — and the American worker will fall further behind.

That is why we are planning to introduce bold legislation to address this crisis. Our bill will prohibit a corporation from buying back its own stock unless it invests in workers and communities first,

including things like paying all workers at least \$15 an hour, providing seven days of paid sick leave, and offering decent pensions and more reliable health benefits.

More recently, Congress passed the Inflation Reduction Act of 2022, which imposed a 1% excise tax on share repurchases to be paid by firms (in addition to the taxes paid by investors who sell their shares back to firms). In his 2023 State of the Union Address, President Biden proposed increasing the excise tax rate on share repurchases to 4%.

The political focus on repurchases plausibly reflects the facts that they now dominate the payout supply and are typically announced as large dollar amounts while dividends are typically described in smaller per-share amounts. Another possible factor is that many members of the general public have embraced basic agency theory and thus find it credible that corporate managers would use share repurchases to further their narrow self-interests. Whatever the underlying explanations, it is clear that politicians and social activists have been able to promote a narrative that vilifies repurchases as a tool managers use to enrich themselves and wealthy shareholders, while ostensibly failing to make socially desirable investments.

Kolhatkar (2019) traces the politicization of buybacks to articles by William Lazonick (2008, 2014), who was Jensen's colleague at Harvard Business School and who, as far back as the 1980s, harbored concerns about Jensen's focus on shareholder value. Kolhatkar details Lazonick's early doubts about the shareholder-value philosophy, and how he eventually came to view share buybacks as promoting income inequality and preventing firms from investing adequately in workers and innovation.

Lazonick (2008) highlighted large share repurchases by major banks right before the Financial Crisis. His key point is that these cash distributions weakened banks and contributed to a need for large-scale equity infusions, including taxpayer-funded bailouts. He concluded:

The crisis in the US financial sector demonstrates that the so-called “free cash flow” distributed as buybacks was not really free.

Although the bailout of large financial institutions clearly put share repurchases on the political radar, political concerns over buybacks increased significantly about 10 years ago, with the publication of Lazonick's (2014) article. That article received the 56th annual McKinsey Award for the most influential

article published in the venerable (and itself highly influential) *Harvard Business Review*. See McKinsey (2015) for a description of the impact of Lazonick's HBR article. As Kolhatkar (2019) reports:

Lawmakers took note. Lazonick told me that, in 2015, he was consulted by Hillary Clinton's campaign – which cited his *Harvard Business Review* article in its economic platform – as well as by Bernie Sanders's campaign. Some Republicans have even started to point to buybacks as the cause of America's political and economic ills. A year ago, one of Senator Marco Rubio's policy advisers contacted Lazonick to express interest in his research. Rubio later published a report called "American Investment in the 21st Century," in which he expressed concern that economic growth was driven more by finance than by actual productivity, and mentioned the buyback phenomenon several times. Senator Elizabeth Warren has made the idea a theme of her campaign. She "picked up the message right away," Lazonick told me. "She said the reason we don't have prosperity for workers is that companies aren't sharing it. That is the critical point."

Prominent politicians, environmental activists, and Lazonick (2008, 2014) have one common theme linking their criticisms of share repurchases. That theme is an attempt to influence corporate resource allocation in ways that they believe would be socially beneficial.

There are two problems with this anti-repurchase narrative. The first is that the narrative is rooted in allegations about share repurchases that are specious or unsupported empirically, as detailed by Edmans (2017), Asness, Hazelkorn, and Richardson (2018), Fried and Wang (2018), and Guest, Kothari, and Venkat (2023).

The second is that this narrative will almost surely fail to accomplish its objective. As discussed by DeAngelo (2023), increasing taxes on share repurchases will backfire at growth-stage firms because it will undermine their ability to raise funding for investment. Nor is raising taxes on payouts likely to work at mature firms that generate positive FCF. The reason is that these firms can direct retained cash to investments in liquid financial assets, which puts a high upper bound on the amount of value managers are willing to sacrifice to avoid taxes on immediate payouts.

These predictions are borne out according to the evidence of Barger et al. (2024), who study how firms react to the Inflation Reduction Act's imposition of a 1% excise tax on net repurchases. They report that firms tend to reduce share repurchases and total payouts and increase their holdings of cash and marketable securities, but do not increase outlays for real investment.

6. Concluding thoughts

Michael Jensen was famously intolerant of research focused on technical minutia, which he often bluntly dismissed in colorful language that can't be repeated here. He was unequivocal and relentless in advising faculty colleagues, PhD students, and authors of papers submitted to the *Journal of Financial Economics* to focus on issues of first-order importance for understanding the real world.

Jensen followed his own advice throughout his career, with his 1986 FCF paper providing a stellar example of how to advance theory in a way that fundamentally alters our understanding of real-world corporate-finance behavior. We emphasize two general principles that emerge from Jensen (1986):

- Payout policy is a primary building block of corporate financial policy and central to the process of value creation by firms.
- Payouts matter through their linkage to capital allocation, with the value generated by a firm driven by the extent to which capital is invested wisely and the fruits of that investment eventually distributed to the suppliers of capital.

In our view, Jensen's (1986) paper would have benefited by including an analysis that takes into account firms for which the value of financial flexibility, as measured for example by cash holdings and untapped debt capacity, is important. A richer dynamic theory of corporate financial policy emerges from considering the tradeoffs inherent in balancing the time-varying advantages of financial flexibility against the disciplinary advantages of having managers operate on a tight financial leash, with significant pressure to distribute rather than retain FCF.

Jensen (1986) focused on the value destruction that arises when managers retain FCF and allocate the funds to investments that further their own self-interests. His analysis provided an explanation for why, in the 1980s, profit-oriented activist shareholders used takeover threats and restructuring pressures to force managers to pay out rather than waste FCF. At a more general level, Jensen's FCF argument provides a convincing explanation for the consistently large magnitude of payouts (and concentration among firms with high earnings) that we observe, as well as for the substantial increases in payouts that have occurred, especially since 2000.

Today, Jensen would likely be concerned about value destruction from the pressures firms face to reduce payouts and invest the retained cash in ways favored by politicians and social activists. More than two decades ago, Jensen (2002, p. 43) presciently anticipated attempts to affect the corporate allocation of capital on the part of special interest groups. Here is what he said (with emphasis added) when making the general case for the advantages of *stockholder* capitalism over *stakeholder* capitalism:

Stakeholder theory plays into the hands of special interests that wish to use the resources of corporations for their own ends. With the widespread failure of centrally planned socialist and communist economies, those who wish to use nonmarket forces to reallocate wealth now see great opportunity in the playing field that stakeholder theory opens to them. Stakeholder theory gives them the appearance of legitimate political access to the sources of decision-making power in organizations, and it deprives those organizations of a principled basis for rejecting those claims. The result is to undermine the foundations of value-seeking behavior that have enabled markets and capitalism to generate wealth and high standards of living worldwide.

These concerns about how stakeholder theory would be used revisit a general theme about political threats to the free-enterprise system that Jensen consistently viewed as a serious problem; see, e.g., Jensen (1976), Jensen and Meckling (1978), and Jensen (1991). It thus does not require a large leap of faith to believe that Jensen would view the recent politicization of corporate payouts as a threat to the efficient functioning of the free-enterprise system and therefore as a research topic worthy of careful attention by corporate-finance specialists.

Figure 1

**Percent of aggregate market capitalization
for firms that distribute cash in each year 1971-2023**

The figure reports the total market capitalization for all firms that distribute cash in the year in question as a percent of the market capitalization of all firms in the full sample. For each firm, market capitalization is the year end value of all outstanding common shares. The full sample consists of firms that (i) are included in the CRSP/Compustat data file in at least one year over 1970-2023, (ii) are incorporated in the U.S., (iii) have a CRSP share codes of 10 or 11, and (iv) are listed on a major exchange (exchange code 1, 2, or 3). Firms in all industries are included in the sample, i.e., we do not exclude financial firms and utilities. Appendix A provides relevant data definitions and describes requirements on specific data availability for a firm to be included in the sample.

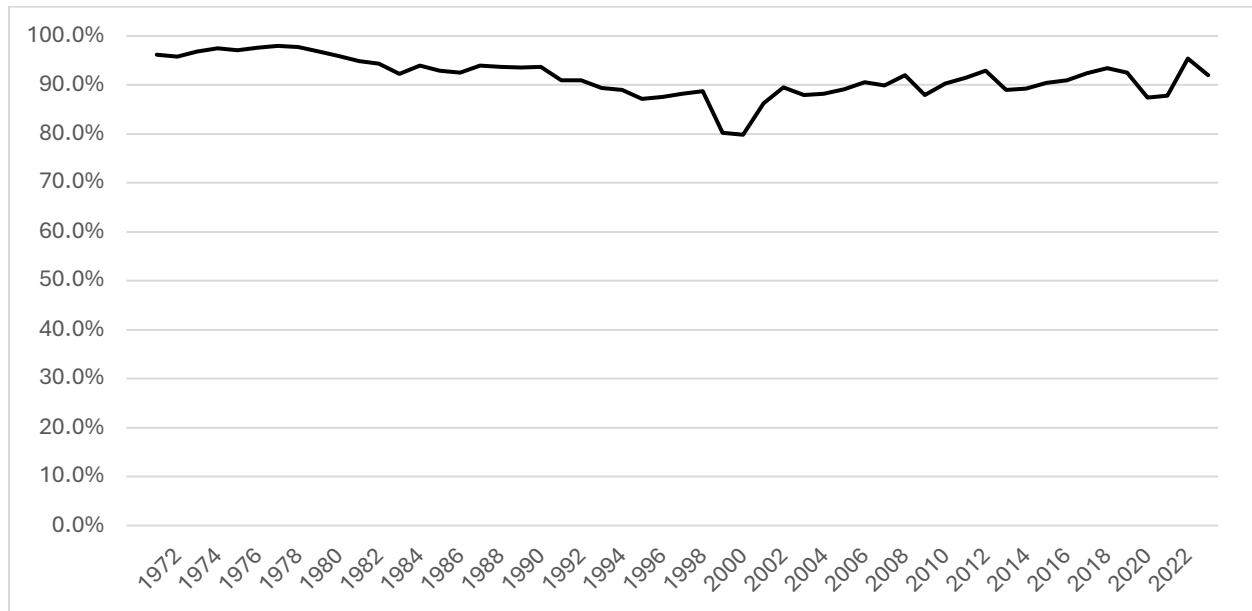
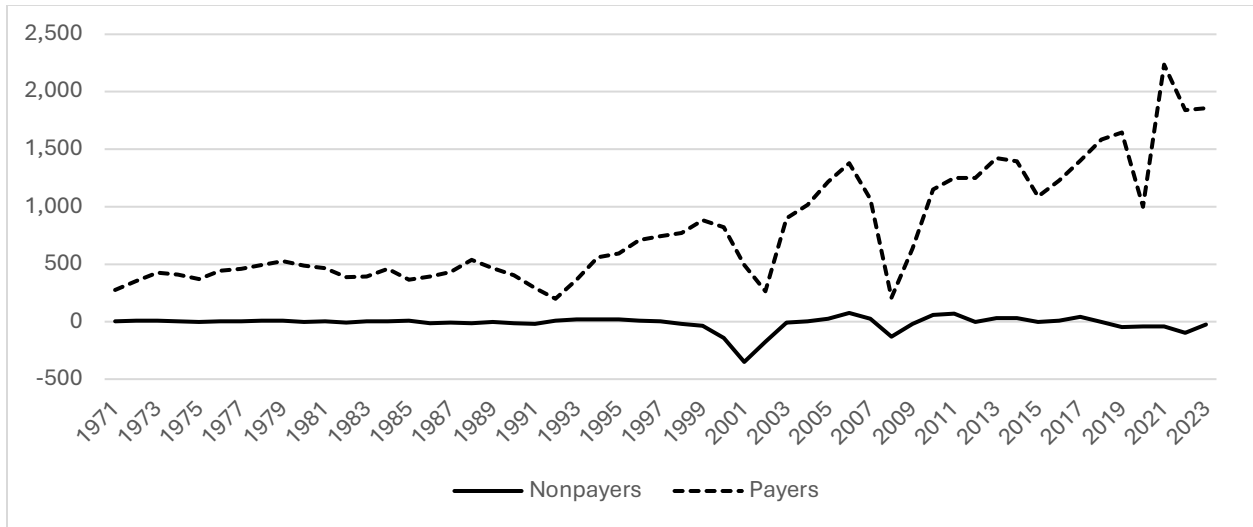


Figure 2

Aggregate real earnings (\$ billions in 2023 dollars) for sample partitioned into firms that pay out and that do not pay out cash: 1971-2023

The dashed plots report total earnings – measured as net income in panel A and as operating income in panel B – for all firms that distribute cash in the year in question, while the solid plots report total earnings for firms that do not distribute cash. The full sample consists of firms that (i) are included in the CRSP/Compustat data file in at least one year over 1970-2023, (ii) are incorporated in the U.S., (iii) have a CRSP share codes of 10 or 11, and (iv) are listed on a major exchange (exchange code 1, 2, or 3). Firms in all industries are included in the sample, i.e., we do not exclude financial firms and utilities. Appendix A provides relevant data definitions and describes requirements on specific data availability for a firm to be included in the sample.

A. Aggregate net income



B. Aggregate operating income

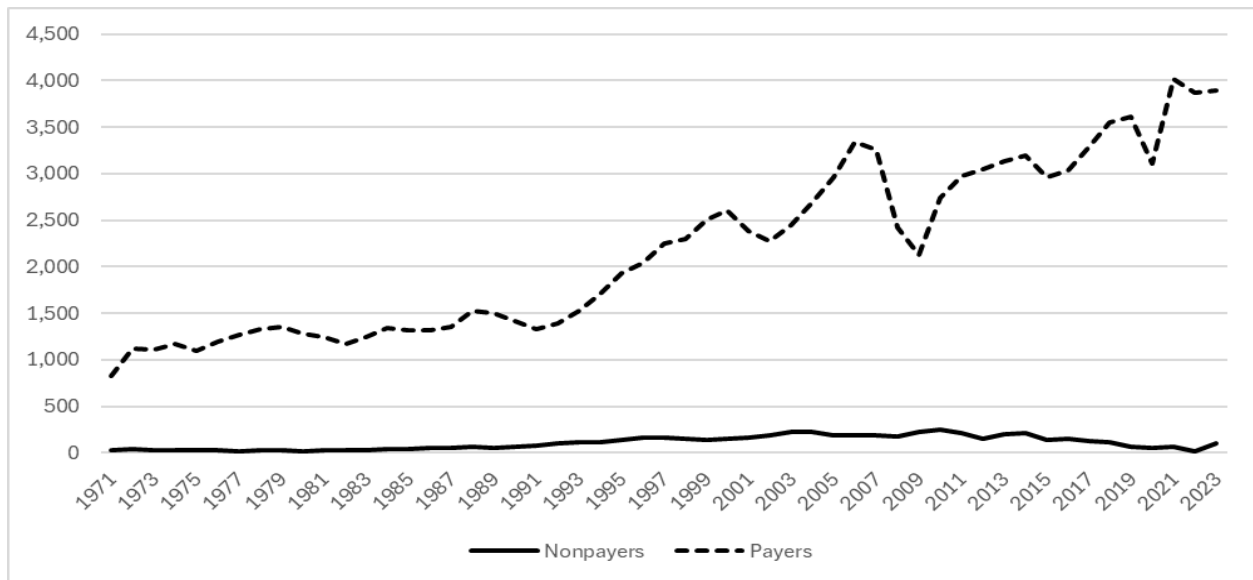


Figure 3

**Aggregate real payouts (\$ billions in 2023 dollars)
and their dividend and share-repurchase components: 1971-2023**

The figure reports the total payouts (dividends plus share repurchases) for all firms that distribute cash in the year in question, partitioned into aggregate dividend and share-repurchase components. All observations are converted to 2023 dollars using the consumer price index. The full sample consists of firms that (i) are included in the CRSP/Compustat data file in at least one year over 1970-2023, (ii) are incorporated in the U.S., (iii) have a CRSP share codes of 10 or 11, and (iv) are listed on a major exchange (exchange code 1, 2, or 3). Firms in all industries are included in the sample, i.e., we do not exclude financial firms and utilities. Appendix A provides relevant data definitions and describes requirements on specific data availability for a firm to be included in the sample.

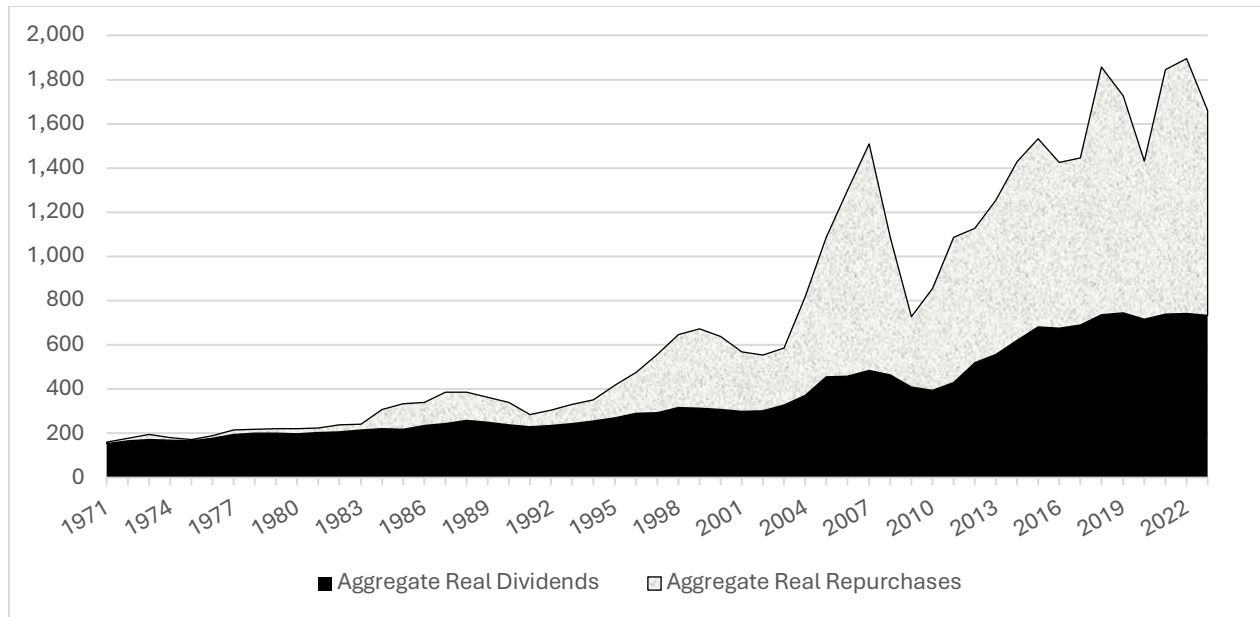


Figure 4

**Aggregate real payouts (\$ billions in 2023 dollars)
partitioned into their main industry components: 1971-2023**

The figure reports the total payouts (dividends plus share repurchases) for all firms that distribute cash in the year in question, partitioned into financial firms, utilities, and the set of nonfinancial and nonutility firms, which we label industrials. All observations are converted to 2023 dollars using the consumer price index. Appendix A provides relevant data definitions and describes requirements on specific data availability for a firm to be included in the sample.

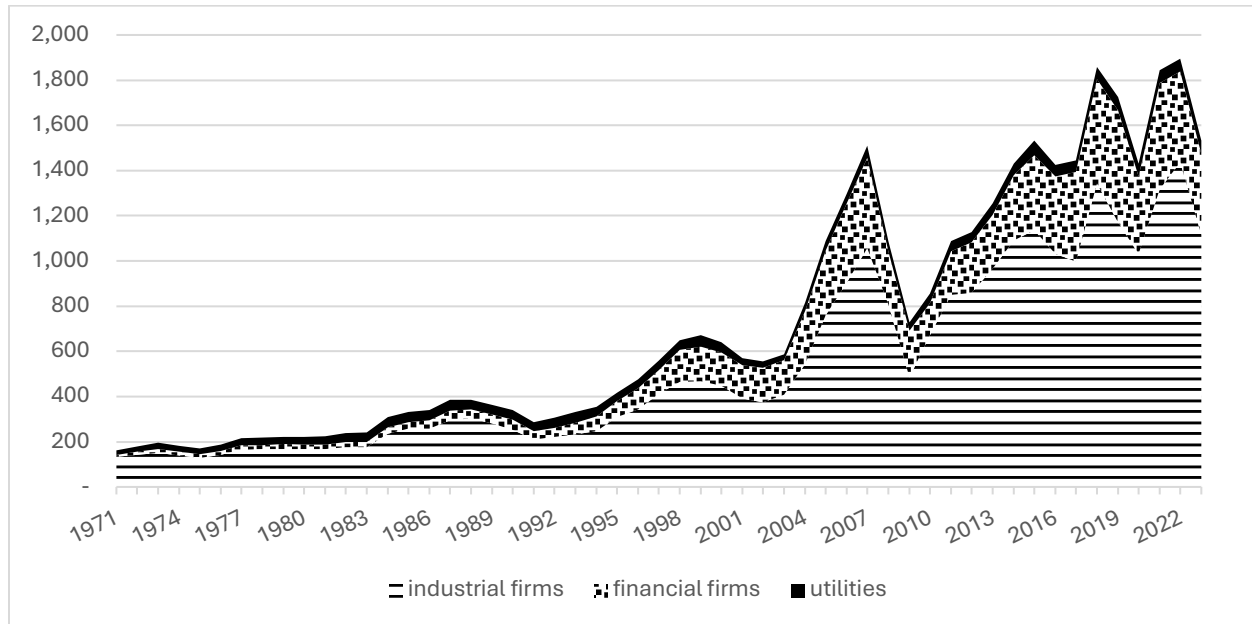
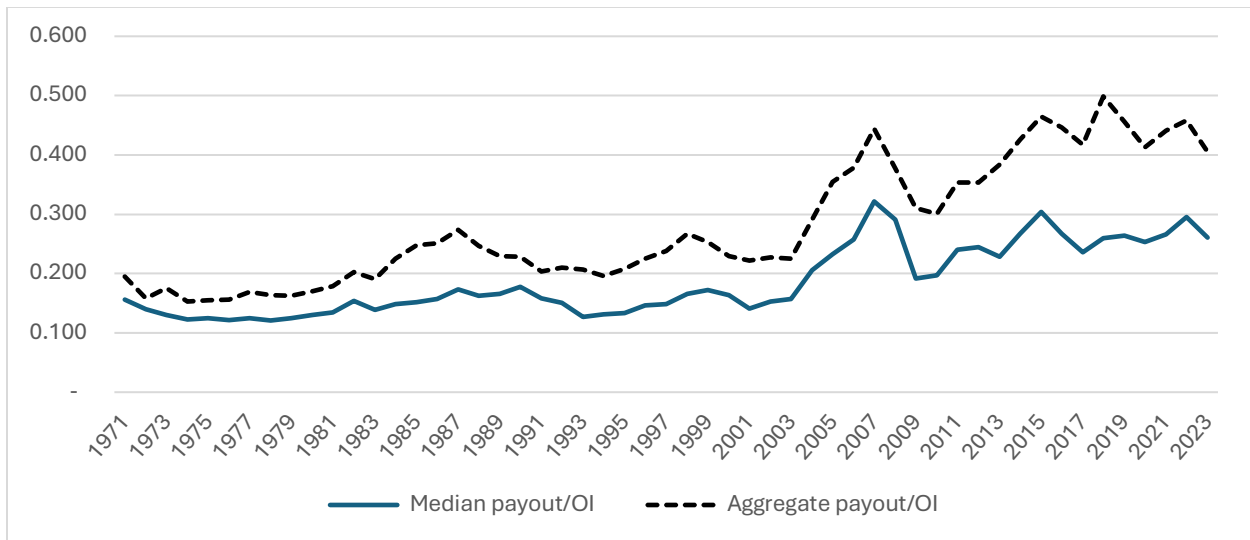


Figure 5

Payouts as a fraction of operating income and net income: 1971-2023

The figure reports aggregate and median-firm payout ratios for firms with positive operating income (OI) in panel A and positive net income (NI) in panel B. The aggregate payout ratio is the sum of all payouts divided by the sum of all operating income or net income in the year in question. The full sample consists of firms that (i) are included in the CRSP/Compustat data file in at least one year over 1970-2023, (ii) are incorporated in the U.S., (iii) have a CRSP share codes of 10 or 11, and (iv) are listed on a major exchange (exchange code 1, 2, or 3). Firms in all industries are included in the sample, i.e., we do not exclude financial firms and utilities. However, the plots for industrial firms are very close to those shown below for the full sample. Appendix A provides relevant data definitions and describes requirements on specific data availability for a firm to be included in the sample.

A. Payouts as a fraction of operating income



B. Payouts as a fraction of net income

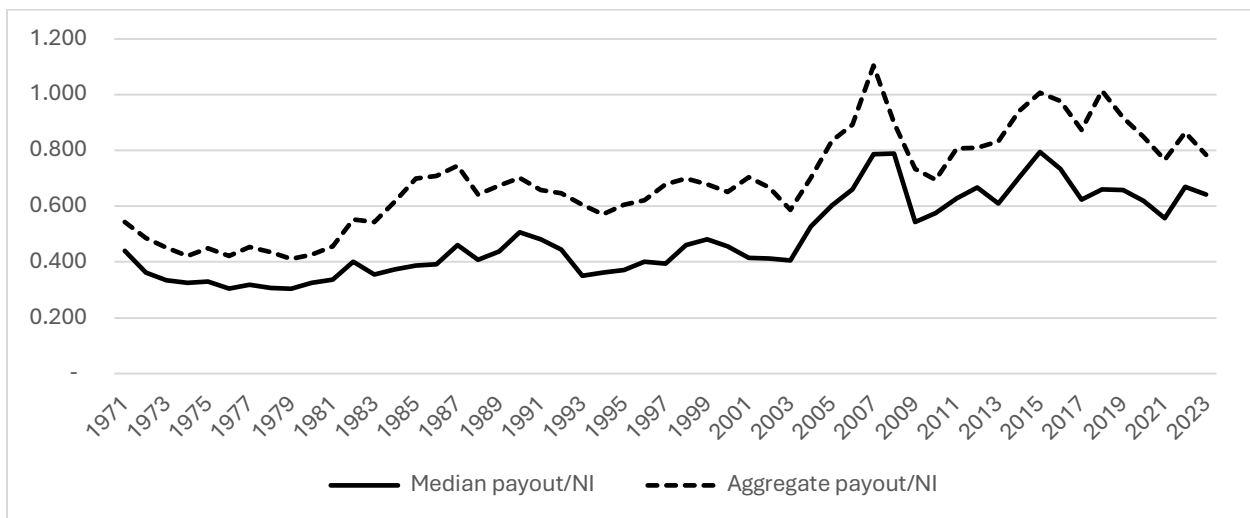


Figure 6

Aggregate real cash balances (\$ billions in 2023 dollars) for sample partitioned into firms that pay out and do not pay out cash: 1971-2023

The figure reports aggregate real holdings of cash plus marketable securities for the full sample of firms partitioned according to whether the firm pays out or does not pay out cash in the year in question. All observations are converted to 2023 dollars using the consumer price index. The full sample consists of firms that (i) are included in the CRSP/Compustat data file in at least one year over 1970-2023, (ii) are incorporated in the U.S., (iii) have a CRSP share codes of 10 or 11, and (iv) are listed on a major exchange (exchange code 1, 2, or 3). Firms in all industries are included in the sample, i.e., we do not exclude financial firms and utilities. Appendix A provides relevant data definitions and describes requirements on specific data availability for a firm to be included in the sample.

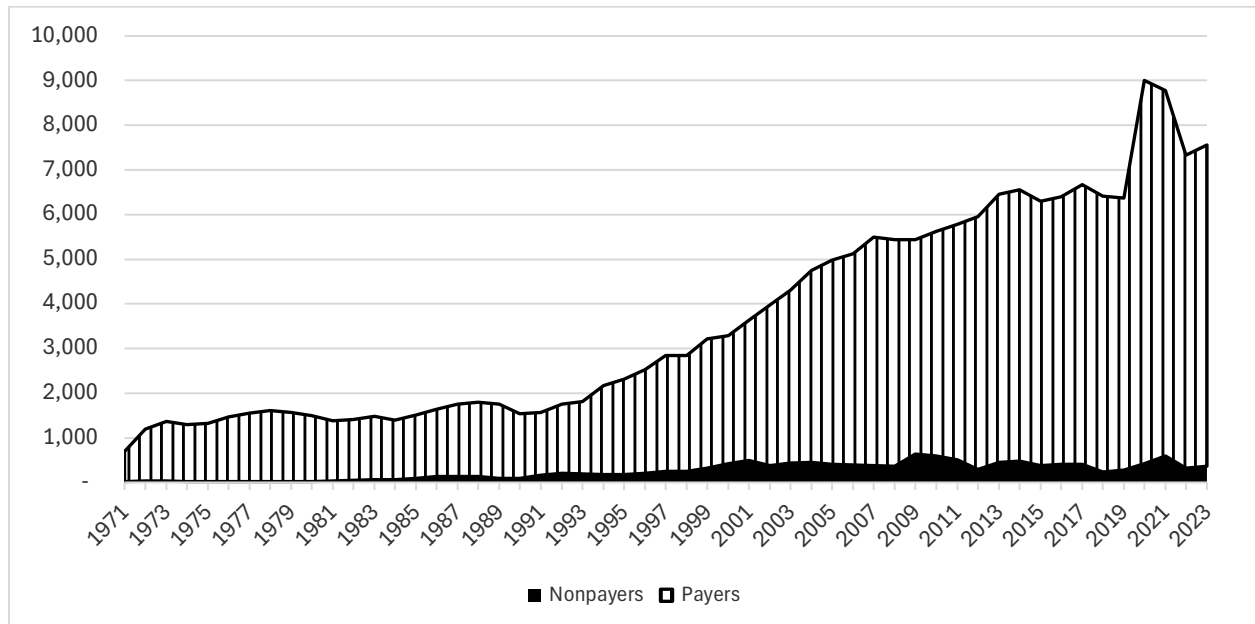


Figure 7

Aggregate and median cash-balance ratios for firms that pay out cash: 1971-2023

The figure reports aggregate and median cash-balance ratios for the full sample of firms that pay out cash in each year over 1971 to 2023. The aggregate cash-balance ratio in a given year equals the sum of the holdings of cash plus marketable securities for all payers divided by sum of the book values of total assets for those firms. The median Cash/TA ratio is the cross-sectional median (for the year in question) across the ratios of cash plus marketable securities for all firms that pay out cash. The full sample consists of firms that (i) are included in the CRSP/Compustat data file in at least one year over 1970-2023, (ii) are incorporated in the U.S., (iii) have a CRSP share codes of 10 or 11, and (iv) are listed on a major exchange (exchange code 1, 2, or 3). Firms in all industries are included in the sample, i.e., we do not exclude financial firms and utilities. Appendix A provides relevant data definitions and describes requirements on specific data availability for a firm to be included in the sample.

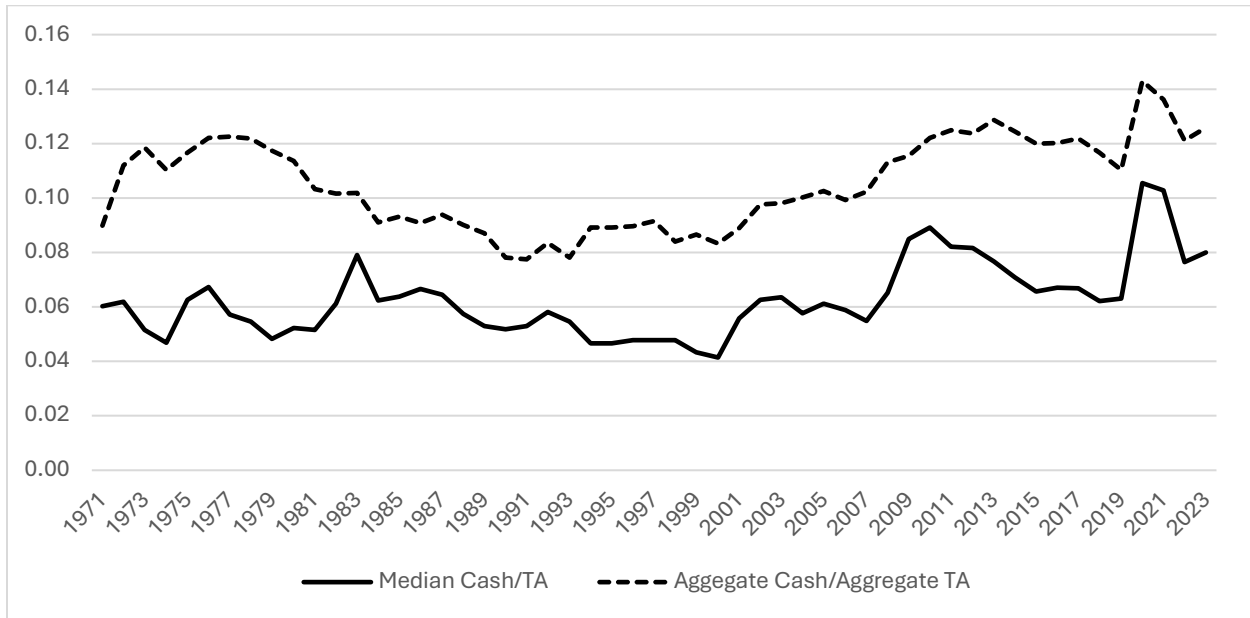


Figure 8

Payouts as a fraction of cash balances: 1971-2023

The figure reports aggregate and median-firm cash-balance-based payout ratios for all sample firms over 1971-2023. The aggregate ratio is the sum of all firms' payouts divided by the sum of (i) all firms' end-of-year (i.e., post-payout) holdings of cash plus marketable securities and (ii) all firms' payouts in the year in question. The sample contains industrial, financial, and utility firms.

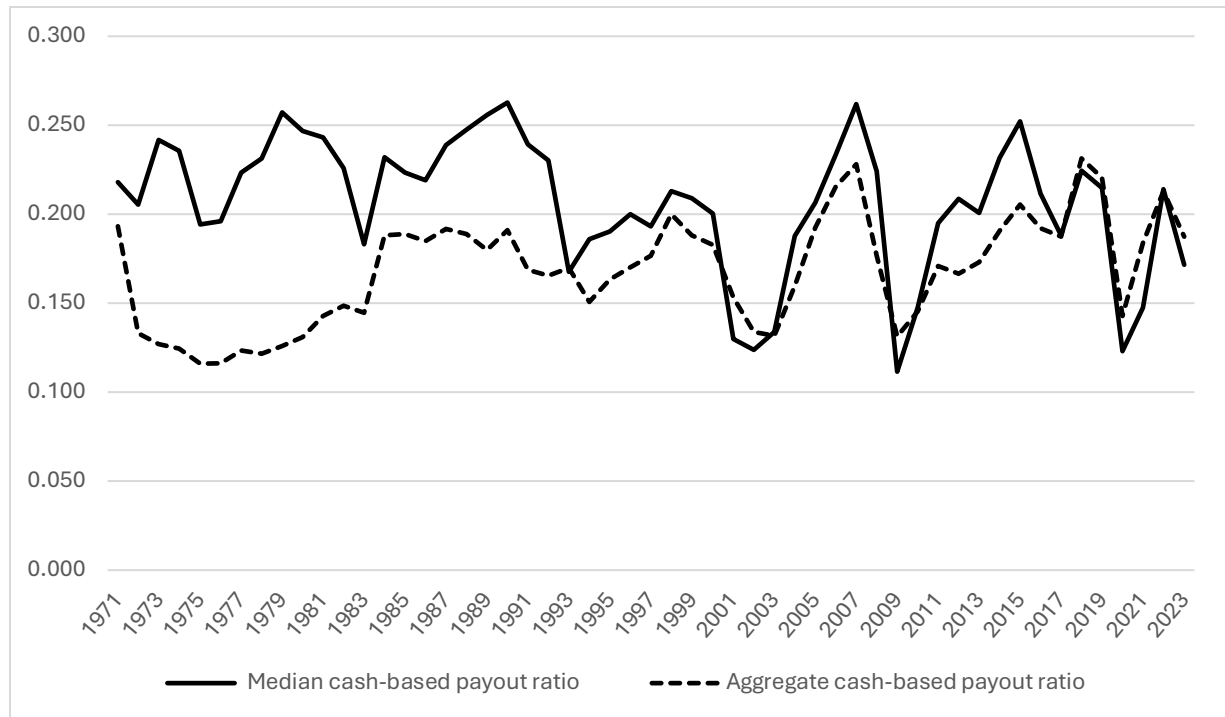


Table 1
Corporate payouts: 1986 versus 2023

In each panel, the full sample for the year in question is partitioned according to the dollar value of the firm's total payout in that year. So, for example, the first row in each panel reports data for the 10 firms that made the 10 largest total payouts, while the next row reports data for the 25 firms that made the 25 largest total payouts. Firms in all industries are included in the sample, i.e., we do not exclude financial firms and utilities. Appendix A provides complete details on sample inclusion criteria.

	1986	2023
Number of sample firms	5,297	4,091
Number of firms that paid out cash	2,904	2,999
Percent of firms that paid out cash	54.8%	73.3%
A. Percent of aggregate payouts made by:		
10 largest payers	19.1%	22.4%
25 largest payers	30.5%	34.9%
100 largest payers	60.4%	48.2%
250 largest payers	79.1%	79.5%
500 largest payers	90.0%	90.0%
All payers pooled together	100.0%	100.0%
B. Percent of aggregate net income from:		
10 largest payers	14.9%	25.0%
25 largest payers	24.0%	36.2%
100 largest payers	49.6%	52.7%
250 largest payers	71.7%	82.5%
500 largest payers	85.6%	92.8%
All payers pooled together	104.1%	101.2%
C. Percent of aggregate market capitalization for:		
10 largest payers	9.9%	20.6%
25 largest payers	17.4%	30.9%
100 largest payers	38.8%	41.3%
250 largest payers	56.3%	66.2%
500 largest payers	71.7%	76.8%
All payers pooled together	92.5%	92.3%
D. Percent of payouts from share repurchases:		
10 largest payers	37.1%	72.9%
25 largest payers	36.6%	70.6%
100 largest payers	36.1%	63.6%
250 largest payers	33.5%	61.5%
500 largest payers	32.0%	56.9%
All payers pooled together	30.5%	55.9%
E. Percent of payers that repurchased shares:		
10 largest payers	70.0%	100.0%
25 largest payers	72.0%	100.0%
100 largest payers	80.0%	93.0%
250 largest payers	68.0%	88.8%
500 largest payers	57.2%	87.6%
All payers pooled together	31.7%	68.3%

(table continued)

F. Percent of payers that paid dividends:		
10 largest payers	100.0%	80.0%
25 largest payers	100.0%	92.0%
100 largest payers	100.0%	88.0%
250 largest payers	99.2%	89.2%
500 largest payers	97.6%	85.8%
All payers pooled together	81.4%	59.1%

Table 2
Top 10 firms ranked by payouts:1986 versus 2023

This table examines the 10 firms that made the largest total (dividend plus share repurchase) cash payouts cumulated over 1982-1986 (panel A), 2019-2023 (panel B), 1986 alone (panel C), and 2023 alone (panel D). Firms in all industries are included in the sample, i.e., we do not exclude financial firms and utilities. Appendix A provides relevant data definitions and describes requirements on specific data availability for a firm to be included in the full sample. In panel C, the symbol * denotes the firm was subject to visible hostile takeover activity. In all panels, an entry with --- indicates the firm is not publicly held during the period in question. Cash/TA denotes the percent equivalent of the ratio of cash plus marketable securities to total assets at year-end 1986. In the second column of Panel A, Oracle's low rank reflects the fact that it repurchased a small amount of stock over 1982-1986 and 4,352 firms distributed cash in at least one year in that interval.

A. 1986: Top 10 firms ranked by their cash payouts cumulated over the five years ending in 1986 (amounts in constant 2023 \$ billions)

Payout rank		Company	Dollar values and ranks for 1982-1986					Cash/TA at year-end 1986
1982- 1986	2019- 2023		Cumulated payout (\$B)	Dividends as a % of payout	Dividend rank	Cumulated net income (\$B)	Net income rank	
1	7	Exxon	58.7	66.1%	2	72.9	2	5.5%
2	---	AT&T	46.4	100.0%	1	32.4	4	6.7%
3	46	IBM	39.8	89.6%	3	81.3	1	12.6%
4	---	Atlantic Richfield	22.6	44.3%	11	12.8	20	11.1%
5	90	General Motors	20.5	91.0%	4	46.9	3	5.5%
6	37	Conoco	19.2	27.1%	37	8.4	37	9.2%
7	---	Amoco	19.1	64.7%	7	25.3	7	1.9%
8	144	General Electric	17.7	76.0%	5	32.0	5	5.5%
9	---	Union Carbide	15.9	38.7%	24	1.9	240	3.9%
10	---	Texaco	15.4	72.2%	9	14.1	17	8.6%

B. 2023: Top 10 firms ranked by their payouts cumulated over the five years ending in 2023 (amounts in constant 2023 \$ billions)

Payout rank		Company	Dollar values and ranks for 2019-2023					Cash/TA at year-end 2023
2019- 2023	1982- 1986		Cumulated payout (\$B)	Dividends as a % of payout	Dividend rank	Cumulated net income (\$B)	Net income rank	
1	913	Apple	537.8	14.9%	3	439.4	1	17.5%
2	---	Alphabet (Google)	280.0	0.0%	---	307.9	4	27.6%
3	---	Microsoft	229.7	40.0%	1	316.9	3	27.0%
4	210	JP Morgan Chase	150.4	47.4%	4	219.5	5	23.2%
5	---	Meta (Facebook)	133.9	0.0%	---	162.8	6	28.5%
6	447	Bank of America	129.5	33.2%	12	144.0	7	19.6%
7	1	Exxon Mobil	116.6	70.2%	2	109.7	9	8.4%
8	364	Wells Fargo	114.2	29.8%	18	83.7	15	16.4%
9	3584	Oracle	91.1	20.6%	37	53.6	26	7.6%
10	13	Chevron	90.9	61.5%	8	72.2	20	3.2%

(table continued)

C. 1986: Top 10 firms ranked by their 1986 total payouts (amounts in constant 2023 \$ billions)

Payout rank		Company	Dollar values and ranks for 1986					Cash/TA at year-end 1986
1986	2023		Payout (2023 \$B)	Dividends as a % of payout	Dividend rank	Net income (2023 \$B)	Net income rank	
1	---	Union Carbide*	12.6	26.4%	7	1.4	65	3.9%
2	54	IBM	11.6	64.5%	1	13.3	2	12.6%
3	4	Exxon	9.4	76.6%	2	14.9	1	5.5%
4	24	General Motors	6.5	71.5%	4	8.2	4	5.5%
5	558	FMC*	5.0	100.0%	3	0.4	257	6.9%
6	---	SmithKline Beckman	4.5	14.2%	74	1.4	60	10.3%
7	---	Colt Industries*	4.3	100.0%	5	0.3	362	5.0%
8	45	General Electric	3.9	75.2%	8	6.9	5	5.5%
9	---	AT&T	3.8	100.0%	6	0.4	275	6.7%
10	58	Ford Motor	3.6	45.8%	27	9.1	3	22.5%

D. 2023: Top 10 firms ranked by their 2023 total payouts (amounts in constant 2023 \$ billions)

Payout rank		Company	Dollar values and ranks for 2023					Cash/TA at year-end 2023
2023	1986		Payout (2023 \$B)	Dividends as a % of payout	Dividend rank	Net income (2023 \$B)	Net income rank	
1	360	Apple	97.7	15.3%	2	96.7	1	17.5%
2	---	Alphabet (Google)	71.3	0.0%	---	73.8	3	27.6%
3	---	Microsoft	42.3	47.1%	1	72.7	4	27.0%
4	3	ExxonMobil	32.7	45.7%	3	36.0	7	8.4%
5	---	Meta (Facebook)	26.8	0.0%	---	39.1	6	28.5%
6	24	Chevron	26.3	43.1%	6	21.4	13	3.2%
7	185	JP Morgan Chase	23.4	57.8%	4	49.6	5	23.2%
8	358	Wells Fargo	19.5	30.4%	24	19.1	14	16.4%
9	12	Johnson & Johnson	16.8	70.0%	5	35.2	8	13.7%
10	63	Procter & Gamble	16.4	55.0%	11	14.7	20	6.8%

Appendix A: Sample composition and share-repurchase measurement issues

Our sample consists of firms that (i) are included in the CRSP/Compustat data file in at least one year over 1971-2023, (ii) are incorporated in the U.S., (iii) have a CRSP share codes of 10 or 11, and (iv) are listed on a major exchange (exchange code 1, 2, or 3). Firms in all industries are included in the sample, i.e., we do not exclude financial firms and utilities. For inclusion of a given firm-year observation in the sample, we require non-missing values of total assets (AT), and market capitalization (CSHO and PRCC_F).

Accurate measures of the dollars spent repurchasing common shares can be difficult to extract from the Compustat file. The literature has an extensive discussion of this issue, and we draw on that discussion for the approach we employ to measure repurchase amounts.

Stephens and Weisbach (1998) examine a variety of measures to estimate the number of shares acquired by a firm, including measures based on decreases in the firm's shares outstanding, dollars spent reacquiring securities, and increases in the dollar value of the firm's treasury stock. Subsequent researchers have attempted to improve the accuracy of these estimates. For example, Jagannathan et al. (2000) examine both monthly decreases in CRSP shares outstanding and purchases of common and preferred stock reported in Compustat. They find that the CRSP measure understates actual repurchases while the Compustat measure overstates them. However, their final sample contains only 35 firms with both CRSP and Compustat data available and includes several outliers, which makes it of limited usefulness.

Because Compustat's purchases of common and preferred stock data item (prstk) includes dollars spent repurchasing preferred stock, Grullon and Michaely (2002) and Kahle (2002) refine this measure by removing changes in the value of preferred stock to obtain a better measure of common stock repurchases.

In 2004, the safe-harbor provision created under Rule 10b-18 began requiring firms to disclose actual repurchases on a quarterly basis. Banyl, Dyl, and Kahle (2008) hand collect data from these new disclosures to examine the accuracy of these various estimates of share repurchases. They note that the use of decreases in shares outstanding can understate repurchases when repurchases and distributions occur in the same month. Although stock issues are relatively infrequent for most firms, the exercise of employee stock options is not, and so the change in the number of outstanding shares will understate repurchases in firms with stock option activity.

Multiple problems plague the treasury stock method. First, as with decreases in shares outstanding, this measure understates repurchases when firms purchase and retire treasury shares in the same year. Second, many companies are prohibited by state corporate laws from maintaining treasury stock accounts. In such cases, companies record repurchases as retirements to common stock accounts (paid-in capital), and so the treasury stock measure misidentifies firms in this situation as having zero repurchases. Although Fama and French (2001) adjust the Stephens and Weisbach (1998) treasury stock measure to account for firms that retire treasury shares, Banyl, Dyl and Kahle (2008) find that Compustat frequently misidentifies the retirement-method firms. Third, treasury stock can increase due to the exercise of stock options even if a firm does not repurchase shares. Finally, Compustat did not start reporting treasury stock until 1982, and so this variable is always unavailable in the early years of our sample.

Because the Compustat repurchase variable (prstk) is the cumulated amount of all security repurchases and retirements during the year, it can overstate repurchases of common stock by including outlays for the retirement or redemption of preferred stock and/or the conversion of preferred into common stock. However, firms occasionally net share repurchases and issuances in their cash flow statements. In such instances, Compustat also reports a netted figure and thus understates repurchases.

Despite these shortcomings, Banyi et al. (2008) find that Compustat purchases of common stock (adjusted for decreases in preferred stock) is the best estimate of actual repurchases.

We conduct an extended check on the adjusted-prstkc method using the 15 years of data that has become available since Banyi et al. (2008) was published. We use the fact that, following the 2004 requirement that firms disclose actual repurchases, Compustat began collecting both the number of shares repurchased during the quarter (cshopq) and the average repurchase price (prcraq). We refer to the product of shares repurchased during the quarter times average repurchase price as *actual repurchases*. We collect these data from 2004 through 2023 and cumulate the quarterly actual repurchase amounts during a given fiscal year to arrive at an annual repurchase amount for each firm-year. We compare this total to the corresponding firm-year estimate obtained from Compustat's purchases of common stock, adjusted for decreases in preferred stock.

Consistent with Banyi et al.'s (2008) finding that the Compustat measure overstates repurchases on average, we find that this measure is 6% higher (in the median year) than the measure obtained using the quarterly data on the number of shares repurchased and the average repurchase price. For a sample of firms with the most extreme differences between these two measures, we directly examine the firm's 10-Qs and 10-Ks.

We find that Compustat's prstkc variable often overstates repurchases by including tax payments related to net share settlements of equity awards and redemptions of common stock due to business combinations.

However, the actual repurchase data from Compustat has its own set of problems. For one, the shares repurchased during the quarter, as reported by Compustat, is sometimes off by a factor of 1,000 due to Compustat recording the number incorrectly. We also find that the average repurchase price recorded in the quarterly Compustat data file can be zero even when a repurchase takes place, resulting in understatements of actual dollars spent on repurchases. This problem can be severe. For example, Apple undertook repurchases in every quarter of fiscal 2022. However, in the first and second quarter, although the shares repurchased is correct in Compustat, the average repurchase price is missing from the file. A general examination of observations with positive shares repurchased combined with missing repurchase prices reveals that this recorded-data problem tends to occur when the repurchase takes place via an accelerated share repurchase that wasn't settled during the quarter.

Because the average difference between the prstkc-based and actual repurchase measures is small and because we are mostly interested in aggregate numbers, we use Compustat purchases of common stock, prstkc, adjusted for decreases in preferred stock, since it is available for a longer time frame.

We also identify firm-year observations in which the adjusted-prstkc measure is negative during the period from 2004 through 2023 that the actual shares repurchased number is available. For observations in which the adjusted-prstkc measure is negative, we examine the distribution of the actual repurchase measure. The number of observations with a negative estimate using the prstkc measure is small (ranging from 61 to 170 in a year). The 90th percentile of the distribution of actual repurchases is less than 1% in each year, and the median of the 90th percentile across all years is 0.12%. The total dollar value of repurchases missed by the adjusted-prstkc measure is always very small relative to aggregate repurchases.

To measure dollars spent on dividends, we use Compustat variable DV, which is cash dividends from the cash flow statement. While these variables are often the same, Compustat's DVC (ordinary/common dividends) can include items such as spinoffs and dividends accrued on restricted stock but not paid. However, DV is sometimes missing when it shouldn't be. Consequently, if DV is missing but DVC is not, we set dividends equal to DVC.

CRSP has its own set of problems with dividend data, which is why we have adopted the common practice in the literature of using the dividend amounts reported by Compustat. Kahle and Williams (2018) note that there are numerous errors in the raw CRSP distribution data. The CRSP problems include missing stock split information, annual and semi-annual dividends that are coded as quarterly dividends, special dividends that are coded as regular dividends, data entry errors, and distributions that are simply missing from the database.

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